

Bachelor Thesis EDAR

EDAR V370A: An Integrative Analysis of Evolutionary Conservation, Positive Selection, and Structural Tolerance

EDAR VARIANT SET:

Group A: Clinically Described Variants (~27)

From ClinVar (Pathogenic / Likely pathogenic). All variants are absent from 1000 Genomes Phase 3 (AF = 0), which is consistent with the rare pathogenic classification.

rsID	Protein	Consequence	ClinVar	POS GRCh38	AF EAS	AF AFR	AF EUR	AF SAS	AF AMR
rs886039348	W374*	stop_gained	Pathogenic/LP	108,897,133	0.000	0.000	0.000	0.000	0.000
rs557166582	R98W	missense_variant	Pathogenic/LP	108,929,262	0.000	0.000	0.000	0.001	0.000
rs121908450	R89H	missense_variant	Pathogenic/LP	108,929,288	0.000	0.000	0.000	0.000	0.000
rs550414401	A26V	missense_variant	Pathogenic/LP	108,930,217	0.000	0.000	0.000	0.000	0.000
rs886039564	R358Q	missense_variant	Pathogenic/LP	108,897,181	0.000	0.000	0.000	0.000	0.000
rs2470612697	C428R	missense_variant	Pathogenic/LP	108,896,972	0.000	0.000	0.000	0.000	0.000
rs121908453	R420Q	missense_variant	Pathogenic/LP	108,896,995	0.000	0.000	0.000	0.000	0.000
rs2470612633	E433*	stop_gained	Pathogenic	108,896,957	0.000	0.000	0.000	0.000	0.000
rs773132518	R25*	stop_gained	Pathogenic	108,930,221	0.000	0.000	0.000	0.000	0.000
rs2105390253	—	splice_acceptor_variant	Pathogenic	108,906,369	0.000	0.000	0.000	0.000	0.000
rs2470612948	Y406*	stop_gained	Pathogenic	108,897,036	0.000	0.000	0.000	0.000	0.000
rs121908452	R358*	stop_gained	Pathogenic	108,897,182	0.000	0.000	0.000	0.000	0.000
rs1696805876	K327*	stop_gained	Pathogenic	108,906,353	0.000	0.000	0.000	0.000	0.000
rs121908456	E354*	stop_gained	Pathogenic	108,897,194	0.000	0.000	0.000	0.000	0.000
rs1432041144	E311*	stop_gained	Pathogenic	108,907,892	0.000	0.000	0.000	0.000	0.000
rs199544410	C301*	stop_gained	Pathogenic	108,907,920	0.000	0.000	0.000	0.000	0.000
rs773885029	W434G	missense_variant	Pathogenic	108,896,954	0.000	0.000	0.000	0.000	0.000
rs1696608702	I431T	missense_variant	Pathogenic	108,896,962	0.000	0.000	0.000	0.000	0.000
rs368841777	—	splice_donor_variant	Pathogenic	108,923,367	0.000	0.000	0.000	0.000	0.000
rs1574368295	—	splice_donor_variant	Pathogenic	108,906,306	0.000	0.000	0.000	0.000	0.000
rs1553444895	—	splice_donor_variant	Pathogenic	108,906,307	0.000	0.000	0.000	0.000	0.000
rs797044437	—	splice_donor_variant	Pathogenic	108,910,459	0.000	0.000	0.000	0.000	0.000
rs1553445945	—	splice_donor_variant	Pathogenic	108,912,677	0.000	0.000	0.000	0.000	0.000
rs797044435	—	splice_donor_variant	Pathogenic	108,930,963	0.000	0.000	0.000	0.000	0.000
rs2470708774	Y136*	stop_gained	Likely pathogenic	108,923,402	0.000	0.000	0.000	0.000	0.000
rs749688157	C31*	stop_gained	Likely pathogenic	108,930,201	0.000	0.000	0.000	0.000	0.000
rs886041005	C428*	stop_gained	Likely pathogenic	108,896,970	0.000	0.000	0.000	0.000	0.000

Group B: Neutral Common Variants (~6)
From gnomAD v4.1.1. with AF > 1% in at least one superpopulation, AF < 85% globally.

rsID	Protein	Consequence	POS GRCh38	AF EAS	AF AFR	AF EUR	AF SAS	AF AMR
rs61761321	M107V	missense_variant	108,929,235	0.090	0.001	0.000	0.002	0.000
rs114808659	G239R	missense_variant	108,910,887	0.000	0.083	0.000	0.000	0.004
rs79648056	S274S	synonymous_variant	108,908,001	0.000	0.082	0.000	0.000	0.004
rs79798733	V246M	missense_variant	108,910,527	0.000	0.029	0.000	0.000	0.000
rs3749099	P290P	synonymous_variant	108,907,953	0.014	0.005	0.009	0.029	0.004
rs146567337	S380R	missense_variant	108,897,116	0.031	0.000	0.000	0.001	0.000

Group C: Positive Control
V370A (rs3827760) selected as primary positive control
And two Linkage Disequilibrium partners identified by plink2 with r² > 0.88

rsID	Protein	Consequence	ClinVar	POS GRCh38	AF EAS	AF AFR	AF EUR	AF SAS	AF AMR	r² V370A
rs3827760	V370A	missense_variant	—	108,896,851	0.873	0.003	0.011	0.013	—	1.000
rs4676213	—	intron_variant	—	108,866,545	0.859	0.003	0.011	0.014	—	0.897
rs72627476	—	intron_variant	—	108,834,368	0.862	0.003	0.011	0.012	—	0.886

